



TAPAN BORUAH

Roll No.: ME24B008

MECHANICAL ENGINEERING

Minor in Computer Science and Engineering

National Institute of Technology , Arunachal Pradesh

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Github | Website

LinkedIn

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	National Institute of Technology, Arunachal Pradesh	8.77 (Current)	2024-Present
B.Tech. Minor	National Institute of Technology, Arunachal Pradesh	8 (Current)	2025-Present
Senior Secondary	CBSE Board	85.0%	2024
Secondary	CBSE Board	92.6%	2022

PROJECTS

- **Sand Carrying Bot** March. 2026
IIT BHU Tech Fest LinkedIn
 - Engineered a load-carrying robotic vehicle using **ESP32 microcontroller** for wireless control.
 - Integrated **FlySky 2.4GHz transmitter-receiver system** for low-latency communication.
 - Designed and fabricated a **custom sand-carrying container chassis** for efficient material transport.
- **Line Follower Robot** March 2026
IIT BHU Tech Fest Github
 - Designed and developed an autonomous line-following robot using microcontroller-based control system.
 - Integrated IR sensor array module for high-precision line detection.
 - Applied PID-based control logic for smooth and stable trajectory tracking.
- **Piston Cylinder Simulation** Jan. 2026
SOLIDWORKS LinkedIn
 - Modeled and simulated piston-cylinder assembly using **SolidWorks**.
 - Conducted kinematic analysis and motion validation.
- **Maze Solver** Oct. 2025
Addovedi Techfest Github
 - Built an autonomous maze-solving robot using **Arduino**.
 - Integrated **ultrasonic** sensing for real-time distance mapping.
 - Implemented wall-following control logic for dynamic path correction.
- **RC Car** Oct. 2025
Addovedi Techfest Github
 - Engineered a high-performance wireless RC vehicle using **ESP32 microcontroller**.
 - Integrated **FlySky transmitter-receiver system** for low-latency remote communication.
 - Implemented high-current motor control using **BTS7960 motor driver module**.
 - Powered drivetrain using **Johnson DC motors** for enhanced torque output.
- **Fire Fighting Robot** Jan. 2025 - May. 2025
DIY Project Github
 - Developed an autonomous fire-detection and suppression system using **Arduino**.
 - Integrated **flame sensors** for real-time hazard identification.
 - Designed a **servo-actuated** directional water delivery system.
 - Controlled a **DC water pump** module for targeted fire extinguishing.

SKILLS

- **Programming:** C, C++, Python, Java
- **Web Development:** MERN Stack (MongoDB, Express, React, Node.js) , CSS , Tailwind CSS
- **Embedded Systems :** Arduino , ESP32
- **CAD:** Solidworks , AutoCAD
- **Tools/Frameworks:** Vs Code , Git and Github , ROS, OpenCV , Gazebo
- **Operating Systems:** Windows
- **Non Technical:** Music (Prabhakar Degree in Tabla)

POSITIONS OF RESPONSIBILITY

- **Web Coordinator / Co-coordinator**, MUSIC Club, NIT Arunachal Pradesh *Dec 2024 - Current*
- **Vice President**, Robotics club, NIT Arunachal Pradesh *Jan 2026 - Current*

ACHIEVEMENTS

- **Battle of Bands** , Secured **2nd** in college Cultural fest (Atulyam) Band competition *2025*
- **Rajya Puraskar**, Bharat Scouts And Guides *2022*
- **Solo Instrumental Winner**, Secured **1st** Position in Regional level KVS competition *2022*

INTERESTS

- Robotics and Autonomous Systems
- Embedded Systems & Microcontroller Programming
- Mechanical Design and Product Development
- Indian Classical Percussion (Tabla)